

Installation Guide

Information about RAPIDA door installation

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1. Introduction

Dear Readers, this guide will help you on

- Safety instructions
- Mounting
- Commissioning
- Maintenance

of the door.

This guide contains important information on safety and proper use of the door system. Always keep the operating instructions close to your door.

2. Conformity

2.1. Guidelines and standards:

The following guidelines and standards were used in the design and production of the this door:

89/106 / EEC Construction Products Directive

2006/42 / EC Machinery Directive

2004/108 / EC EMC Directive

2006/95 / EC Low Voltage Directive

DIN EN 13241-1 Door product standard

DIN EN 12604 Mechanical aspects, requirements

DIN EN 12453 Safe use of power - operated doors, requirements

Note: The door has been manufactured and tested according to the standards and guidelines listed.

Identification

CE The roller door complies with the applicable standards and guidelines. Conformity has been demonstrated; the corresponding documents are deposited with the manufacturer.

Note: The manufacturers and the declaration of conformity can be found in the appendix of this operating manual.

3. User Notes

3.1. Purpose of this document

This operating manual describes the operation and maintenance of the roller door. It provides important information for safety and efficient handling.

3.2. Safety instructions

Safety instructions are indicated by the pictograms and signal word. The signal word describes the severity of the imminent danger.

	Danger	Immediately threatening danger to the health and life of persons (serious injury or death)
	Warning	Possibly imminent danger to the health and life of persons (serious injury or death)
	Attention	Possibly dangerous situation. (Minor injuries or property damage)

	Bid	Commitment to a particular behavior or activity for the safe handling of the gate.

Warning for specific hazards

		
Danger by electrical tension	Electrical shocking	Danger by lifted loads

3.2.1. To the safety instructions

The safety instructions in this operating manual must be observed and must be followed strictly.

3.3. Warranty and Liability

The warranty claims require proper operation and handling. The manufacturer assumes that all parts at the time of delivery are fault-free in terms of material and workmanship.

In general, our "General Terms and Conditions of Sale and Delivery" shall apply. These have been available for the operator at the latest since the conclusion of the contract. Warranty and liability claims for personal injury and material damage and are excluded if they are attributable to one or more of the following causes:

- Not using the door properly
- Improper installation, commissioning, operation and maintenance of the door
- Door is used in the event of a defective safety device, a safety and protective device that is not properly installed or not functioning
- Do not pay attention to the instructions in the operating instructions regarding the transportation, storage, installation, commissioning, operation, maintenance and setting up of the door
- Unauthorized structural changes at the door
- Unauthorized modification on the drive
- Defective monitoring of machine parts subject to wear
- Improperly performed repairs
- Disasters caused by foreign bodies and force majeure

Disclaimer: The manufacturer assumes no liability for damage caused by operating errors or non-observance of the operating instructions or lack of maintenance and care.

3.4. Obligation of the buyer

Observe the notes in the operating instructions: The basic prerequisites for safe handling and trouble-free operation of this door are the knowledge of the basic safety instructions and the safety regulations. These operating instructions, in particular the safety instructions, must be observed by all people working on the system. In addition, the rules and regulations for accident prevention applicable to the place of use must be observed. The buyer must ensure that the personnel

responsible for the on going maintenance and care of the door are instructed in the operation and all safety-relevant aspects of the door. The instructors must create and keep a record of the instructors.
Exclusion of Liability: The manufacturer accepts no liability for damages resulting from a failure by the purchaser to comply with the instructions. The buyer shall also be responsible for:

- the operating personnel of the door
- turn to the door is available

3.5. Allowed Users

3.5.1. Experts

A qualified person is designated as

- who on the basis of their professional training and experience, have knowledge of the installation to be tested in the field of windows and doors,
- whom are familiar with the relevant state health and safety regulations, directives and generally accepted rules of technology (eg VDE regulations, DIN standards)

so that they can assess the safe working condition of the respective installation. Experts must be required to make their assessment objectively, from the point of view of safety, unaffected by operational or economic circumstances.

3.5.2. Lay people

As a layman, who is not a specialist. The operator must ensure that lay persons are instructed about the operation of the door. They may not perform any activities in the field of assembly, commissioning, maintenance or disassembly work.

3.6. Intended use

The roller door is used in the access area of people and mainly for use as safe access for goods and vehicles, accompanied by persons, in industrial, commercial or private facilities. The door is only:

- for load-free opening and closing of its door leaf.
- to be used in a safety-free condition

Note: For use in environments with aggressive influences (eg sewage treatment plant, laundry, etc.) please consult the manufacturer.



Important: When used in environments with aggressive dust, (eg cement works,

grinders, foundries, etc.) it is important to ensure that no dust can be deposited on the door leaf which leads to increased wear.

The intended use also includes:

- Observe all notes in the operating manual
- Compliance with maintenance and service work

3.7. Improper use

Improper or non-intended use of the door can endanger the user's body and life, inflict injury to the user and damage the door or other property.

Disclaimer: The manufacturer accepts no liability for damages resulting from improper use.

4. General safety instructions

4.1. Proper use

The operational safety of the door is only guaranteed when used as intended.

Proper use is only possible if the door is used exclusively as a vertically installed industrial door. It is only suitable for mounting on load-bearing constructions. Self-assembly must be carried out by an expert.

Furthermore, points that must pay atttention are,

- conditions laid down in the order confirmation
- to the "Technical data" chapter
- required operating instructions in the safety, operating, maintenance and maintenance conditions

If the door is used differently, an "improper use" exists. This can result in property damage and personal injury. The manufacturer is not liable for damages resulting from this. The risk is borne by the user alone.

Storage of the door

The door must not be stored in an open place and therefore not exposed to any moisture, wetness or dirt.

Modifications or alterations

The mechanical and electromagnetic behavior of the door can be impaired by additions or changes of any kind. Therefore, do not make any changes. Only use original parts or parts approved by the manufacturer. These are checked and meet all requirements. The use of parts not approved by the manufacturer can lead to damage. The manufacturer assumes no liability for this.

4.2. Risk of handling the door

When using the door, dangers and impairments may occur

- for the life and body of operators or third parties, and
- for the door itself or other property values.

The basis for safe handling and the trouble-free operation of the door is the knowledge of the safety and user instructions in this manual.

Please note:

- The resistance to the wind load always refers to a closed gate
- The operation of the door can be dangerous!

4.3. Training of personnel

Installation and electrical connection of the door may only be carried out by qualified personnel. These personnel must be trained by Arde and have all the necessary knowledge.

4.4. Safety and protective equipment

The following security and protection devices are installed at the door:

- Main switch or power connector for disconnecting from mains.
- Emergency stop button.
- The safety light curtain. It is used to stop the door when the obstacle or person is detected and to raise it again. In addition, the door opens when the system fails.



Operate the door only when all protective and safety devices are fully installed and functioning!

Protective devices

Faulty or disassembled safety devices can lead to hazardous situations. For that reason

- immediately switch off the system,
- secure against restarting,
- if necessary, switch off the power supply.

Check the safety and protective equipment

Periodically check all safety and protective equipment. See point 11.2.1.

Safety device	Test interval
Emergency stop button (if applicable)	Once a week
Light curtain	Once a week

4.5. Symbols on the door

Warning signs are attached to the door. These signs warn for residual risks, which can not be eliminated by constructive means. Please observe these safety instructions.

Sembol	Description
	Danger by electrical voltage Work on the electrical system only by an electric specialist

5. Transportation

	**Warning!** Raised or floating loads can result in serious injuries or high damage to property! Never stand or pass through under suspended loads.

5.1. Unboxing the door

Improper unboxing of the door packaging can damage the door. Do not cut the packaging with sharp objects.

6. Installation

Arde high-speed roller doors are delivered ready for installation. Depending on the local conditions, the mounting can be done by screwing or doweling. Holes are provided in the brackets and the guide rails for drilling. Arde doors are only suitable for mounting on load-bearing constructions. **Self-assembly must be carried out by an expert.**

	**Danger due to electrical voltage** The electrical connection may only be carried out by a qualified electrician according to the wiring diagram. Observe the valid standards and directives at the installation site.

	**Risk of fatal injuries from crashing doors** Sloping doors from low altitudes can lead to fatal injuries or damage to the property. The door must be securely and permanently absorbed by the fixings. Check the mounting surface for suitability for installation. Maintain all requirements for a secure attachment.

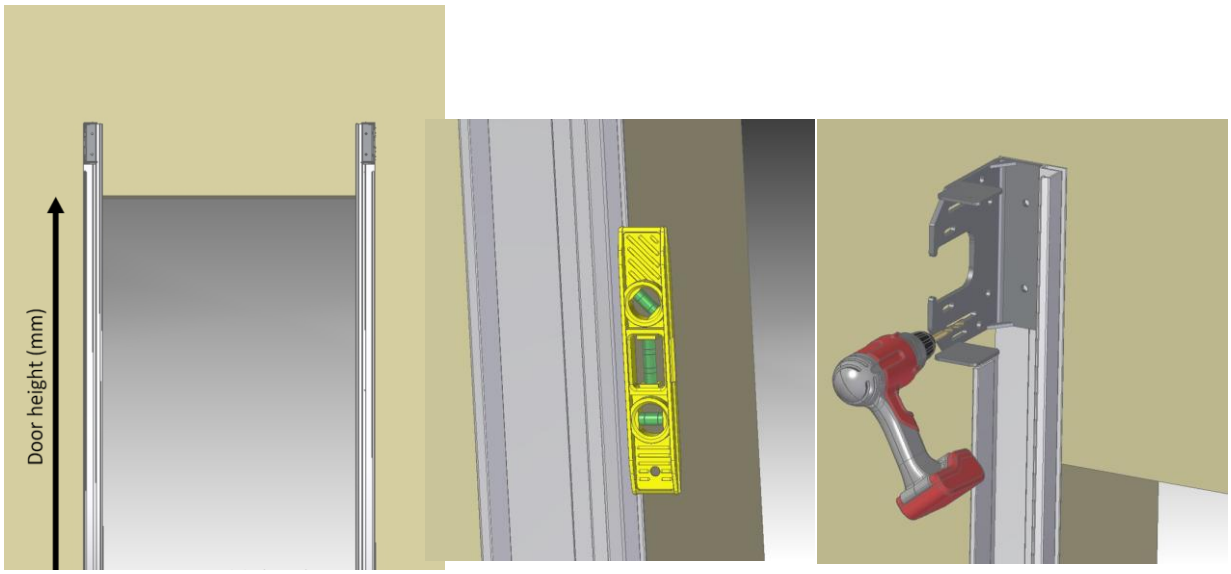
6.1. Claims on electrical products not included in the scope of supply

Electrical products that are not included in the scope of supply must be left for the gateway. They may only be installed by qualified electricians.

	The floor must have the same level in the area of the guide rails. Check the clear passage dimensions of the opening with the order. Pay attention to torsion-free mounting of the guide rails.

6.2. Installation of guide rails

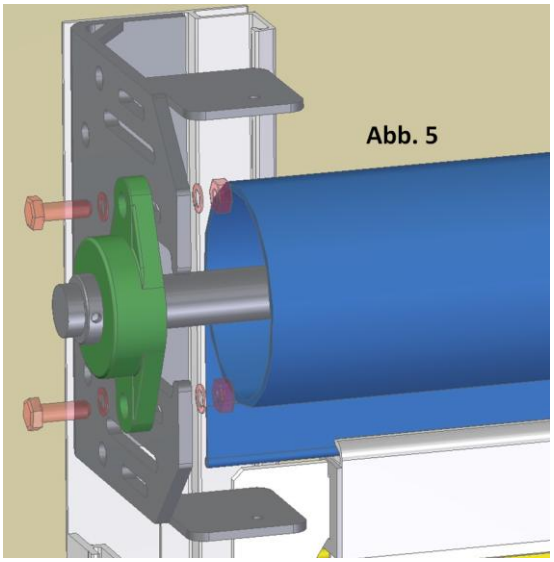
- Check the floor for leveling the guide rails for same level
- Remove the guide rails cover
- Adjust the guide rails if necessary to compensate any height differences
- Set the guide rails to opening dimension and temporary fix them
- Check the openning dimensions again
- Fix the guide rails. Pay attention to sufficient fastening particularly in the area of the brackets
- Fasten the cable tube between the assembled guide rails to the lintel

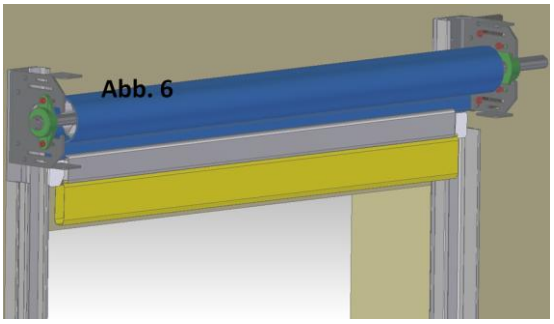


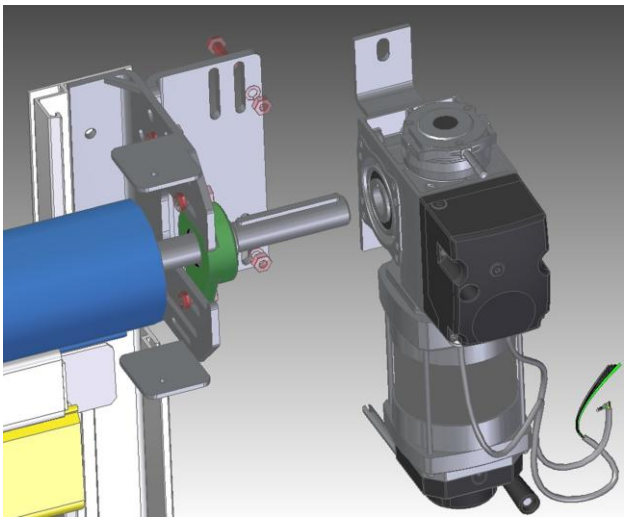
6.3. Mounting the winding shaft

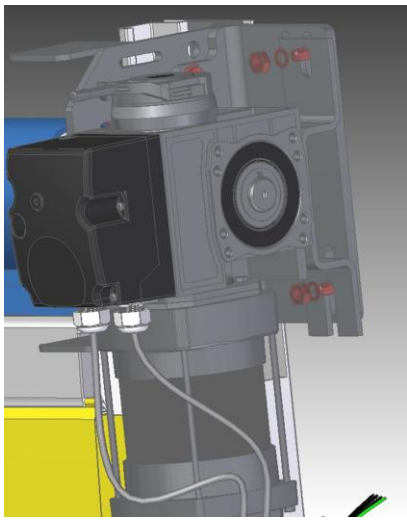
	Pay attention that the curtain should rest between the konsoles while mounting the winding shaft.

- Fasten the flange bearings as shown in Fig. 5 to the brackets using the supplied M10 x 45 hexagon bolts and nuts / washers
- Make sure that both flange bearings are in the same position in the console.
- Make sure that the curtain is centered in the feed hopper of the guide rails.
- Check that the screws to the console are tightly tightened.
- Check that grubscrews of shaft bearings are tightly tightened.
- If necessary, use a lifting device to attach the drive to the shaft stub.
- Attach the drive to the torque support as shown in figures 7 and 8.



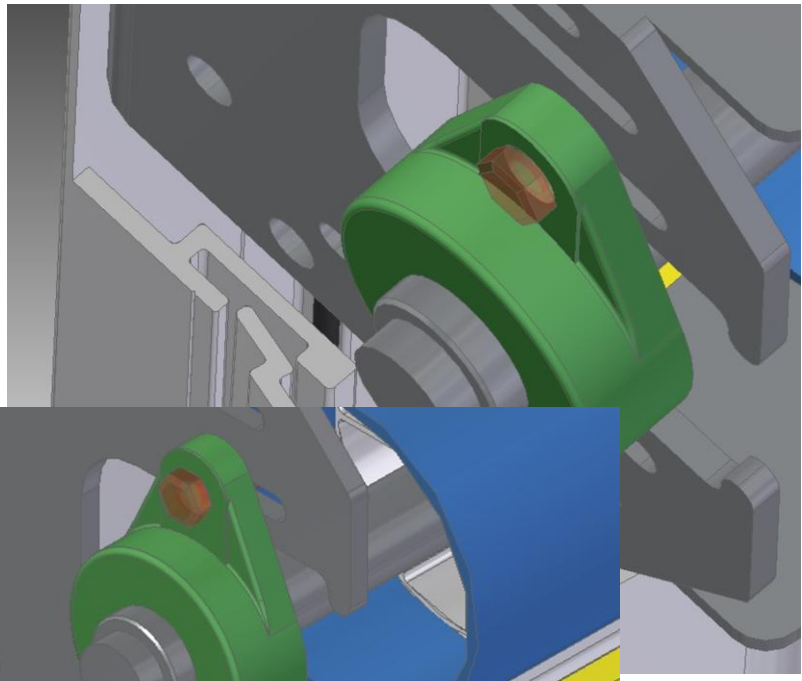






6.4. Assembly of the guide rail cover

- Insert the cover profile of the guide rail into the profile of the wall guide rail provided for this purpose. (See Fig. 9)
- Fasten the cover profile with countersunk head screw M6 x 12mm (See fig. 10)



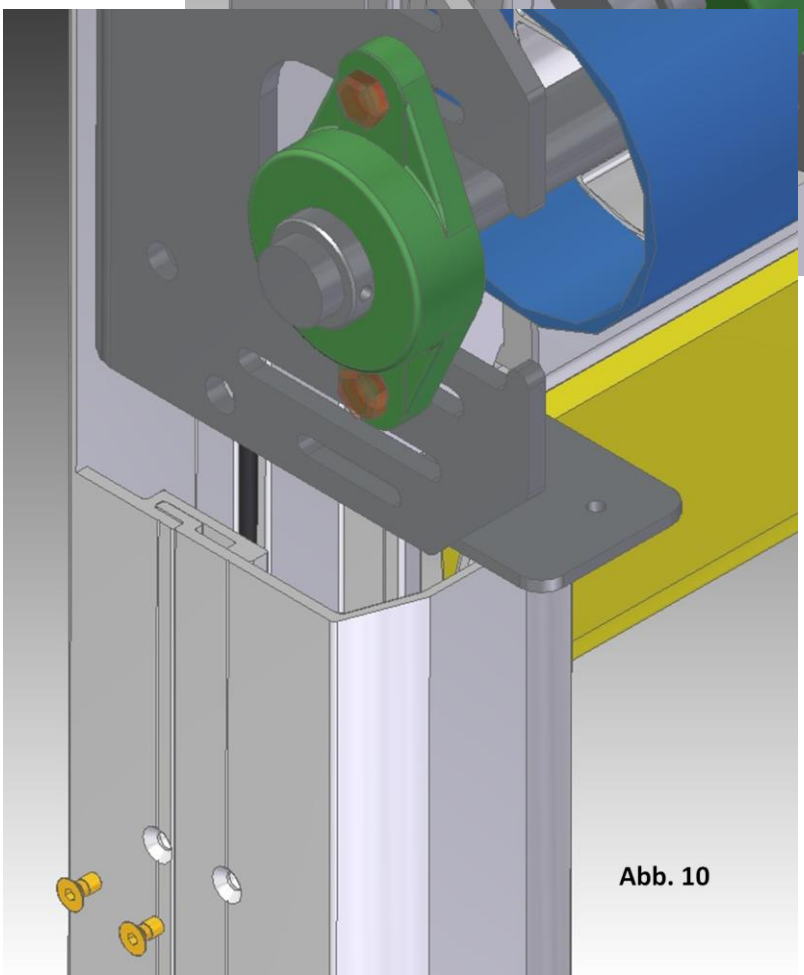


Abb. 10

6.5. Mounting the control unit

6.5.1. Safety instructions for control unit



IMPORTANT INSTRUCTIONS FOR SAFE MOUNTING

Observe all instructions, incorrect assembly can lead to serious injuries!

- The system must be disconnected from the power supply during assembly of the control unit
- The control unit can only be opened if the supply is switched off at all poles. It is not permitted to switch on or operate the control unit in the open state
- All supply circuits must be switched off before accessing the connection terminals
- Before installation, check the control unit for possible transport or other damages. Damages to the interior of the control unit can lead to considerable consequential damage to the control system up to the risk to the user.
- It is forbidden to mount the control unit with a damaged keypads or defective viewing window. Damaged keypads and windows should be replaced.



WARNING

- It is forbidden to touch the electronic parts, especially the parts of the processor circuit. Electronic components can be damaged or destroyed by electrostatic discharge
- Before opening the housing cover of the controller, it must be ensured that there are no drill shavings around, or anything lying on the covert hat might fall into the housing interior.
- Ensure that the controller is mounted without mechanical stress.
- Unused cable entries must be closed by suitable measures to ensure protection class IP54 of the housing.

- The cable entries must not be subjected to any mechanical load, in particular tensile loads.
- Operation on the control system when the CEE connector is removed, is only permissible if the mains supply can be disconnected from the control system by a suitable switch. The power plug or the replacement switch must be easily accessible.
- If the connection cable of this device is damaged, it must be replaced by the manufacturer or its customer service or a similarly qualified person in order to avoid hazards (according to connection type Y according to EN 60335-1).
- When the door is operated in dead-man mode, make sure that the door area can be viewed by the operator. In this operating mode, safety devices such as the safety strip and light barrier may not be effective. If this is not possible for structural reasons, it must be ensured that this operating mode is accessible only by trained personnel in this regard or the function is completely deactivated.
- To avoid damaging the keypad, an operation with pointed objects is prohibited. The keypad is only intended for finger actuation.
- Depending on the design of the door, it may be necessary that the door is operated only with visual contact to the door. In such cases, no remote controller (eg radio) may be used as pulse generator.
- It is imperative to install the control unit with the enclosed wall spacers in order to ensure sufficient heat dissipation of the output stage.
- The controller must not be mounted on flammable substrates (eg wood ≤ 2 mm thick) or in environments with highly flammable substances (carpentry).
- When the controller is installed in an additional housing, e.g. in a barrier housing, case must be taken to ensure sufficient air volume around the control system. This must be at least 0.02 m^3.
- If cooling is not sufficient, an additional heat sink can be installed between the control housing and the additional housing, which then dissipates the heat to the outside (see chapter 4.2 Additional cooler for small housing).

6.5.2. Space required for controller unit



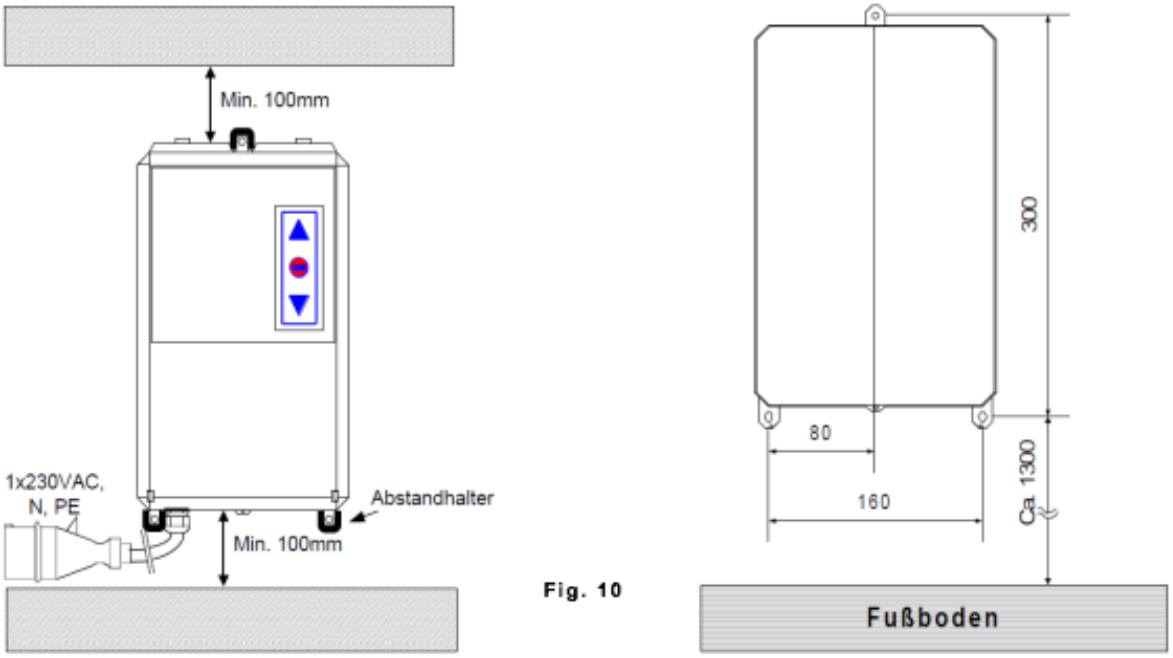


Fig. 10

6.5.3. Electrical connection

ATTENTION

- Connection, testing and maintenance work on the open control unit may only be carried out in a voltage-free state. Particular attention should be paid to the points listed in the "Safety Instructions" section.
- It is not permitted to operate the controller in open state.
- When the control unit is switched off, dangerous voltages remain for up to 5 minutes.
-
- No galvanic isolation during the standstill period between the output stage module and the motor connection terminal.

- Touching the electronic parts is dangerous due to residual voltages.
- The control unit must never be operated with the housing cover open.
- After the installation is completed, check the system if it is set up correctly and the safety system is working properly.
- The control unit can only be opened if the supply is switched off. It is not permitted to switch on or operate the controller in the open state.
- Operation of the controller without connected protective conductor is not permissible. When the protective conductor is not connected, dangerously high voltages occur at the control housing, due to leakage capacitances. Due to the radio interference suppression filters integrated in the control, an increased leakage current up to max. 7 mA (see DIN EN 60335-1 section 16.2). A corresponding piece check was carried out before the device was delivered by the manufacturer.
- Even after the power supply has been switched off, dangerous voltages remain on the intermediate circuit capacitors for up to 5 minutes. The discharge time up to voltage values below 60 VDC is a maximum of 5 minutes. Touching internal control parts within this discharge time is dangerous.
- In the case of a defective switching power supply, the discharge time of the intermediate circuit capacitors can considerably extend to a voltage value below 60 VDC. Discharge times up to 10 minutes can occur.
- Areas of the processor circuit are directly connected to the power supply. If monitoring measurements are to be carried out, it is imperative that no measuring instruments with a PE reference of the measuring circuit are used for measurements in this range of the processor circuit.
- Operation of the control unit with a damaged keypad or viewing window is prohibited. Damaged keypads and windows should be replaced. To avoid damaging the keypad, an operation with pointed objects is prohibited. The keypad is only intended for finger actuation.
- If the potential-free contacts of the relay outputs or other terminal points are externally fed, i. is operated with a dangerous voltage, which can still be present after switching off the control unit, or when the mains plug is disconnected, a corresponding warning sticker must be clearly visible on the control housing.

- When the door is operated in dead-man mode, it must be ensured that the door area can be viewed by the operator since in this operating mode safety devices such as the safety strip and the light barrier do not take effect.
- The setting of the parameters including the speed and the function of all safety devices must be checked. The setting of the parameters, bridges and other controls may only be carried out by trained personnel.
- Before the control unit is switched on for the first time, check that all motor connections on the control unit and motor side are tightened and that the motor is correctly connected to the star or delta. Loose motor connections usually lead to damage to the inverter.
- In the case of short-circuited or extremely overloaded 24 V control voltage, the switching power supply does not operate even though the intermediate circuit capacitors are charged. The ads remain dark. A start-up of the power supply is only possible after elimination of the short circuit or the extreme overload.
- Only shielded, separate motor cables may be used to comply with the EMC directives. The shield must be connected on both sides (motor and control unit side) and no further connections may be routed in the cable. Maximum cable length: 30 m.
- Particularly with fast-running folientors, very high electrostatic charges occur. The discharge of this voltage can lead to damage to the control system. Therefore, appropriate countermeasures must be taken to prevent electrostatic charge.
- It is not permissible to switch on or operate a concrete controller. It can lead to the destruction of the control system.
- Before switching the control supply on for the first time, make sure that the evaluation boards (plug-in modules) are plugged in the correct position. If the cards are stuck or twisted, they can lead to damage to the controller, as well as to the installation of non-approved third-party products.
- Max. Connection cross sections of the circuit board terminals:

	Inducing (starr)	Stranded (w/o end sleeve)	Max. tightening torque [Nm]

Engine clamps	2,5	2,5	0,5
Mains connection	2,5	1,5	0,5
Screw clamps	2,5	1,5	0,5
Plug ins (raster 5 mm)	1,5	1,0	0,4
Plug ins (raster 3,5 mm)	1,5	1,0	0,25

7. Operation

7.1. Procedures

- For doors with automatic closing time, a new UP impulse must be given before each pass.
- The door must only be passed when the door is fully open.
- The door should be passed quickly.
- The door area must be clear when anyone is approaching. No objects or vehicles may be parked in the door area.

	Pulse generators (sensors) with variable settings must be adapted (adjusted) by the customer to the operating sequence. This also applies to ramp hedges, photoelectric sensors, beam detectors, radar detectors, infrared detectors and optical monitoring systems. The settings must be checked by the operator on a monthly basis and, if necessary, adjusted and recalibrated.

7.2. Turn off the gate in case of emergency

Depending on the version, immediate emergency shutdown is possible. For this, the emergency stop button on the control unit must be pressed. When this button is pressed, the door stops immediately.

Danger of serious injury or fatal electric shock



Protect yourself from unauthorized activation of the main switch. For your safety, close the main switch with a lock.



Protect yourself from unauthorized plugging of the power plug. For your safety, attach a sign with the appropriate note.

7.3. Manual override with hand crank

- Insert the hand crank with pressure into the opening at the bottom of the drive and turn it until it clicks into place. The control voltage is thereby interrupted and the door can no longer be operated electrically.
- Turn the hand crank tp open or to close the door.
- Pull out the hand crank, the control voltage is thereby switched on again and the door can be operated electrically.

7.4. Check the safety and protective equipment

For your safety, regularly check the following safety devices.

Check once a week;

- Visual inspection for damage
- The emergency stop button (if present)

Operation;

- Press the emergency stop button during door movement

  => The door must stop immediately.

If the door does not stop, turn off the control unit and secure against restarting. Immediately call service personnel.

The main closing edge;

Operation;

- During the door closing movement, interrupt the monitoring field of the light curtain

  => The door must stop immediately and move upwards.

If the door does not stop, turn off the control unit and secure against restarting. Immediately call service personnel.

7.5. Cleaning

	Switch off the power for the door and close the track traffic before cleaning.

- Clean the curtain regularly. Use only approved PVC cleaners.
- Maintenance work may only be carried out by persons commissioned by the operator.
- The door area should always be kept clean.
- Electrical parts, drive, safety strip etc. must not be exposed to water jets or steam cleaning equipment.

8. Maintenance and service

In any mulfunction call trained service personnel or the customer service of the manufacturer.



Danger due to electrical voltage

Work on the electrical system of the door may only be carried out by a qualified electrician. Before starting to work, switch the door free of stress and secure it against being switched on again.



Crushing

Moving parts can detect and squeeze body parts. Keep unauthorized persons away from the moving door.